

Experiment :- 1.3

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Section/Group: 23BCC-1 (A)

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Subject Name : Building & Release Management

Subject Code : 23CSH-395

1. **Aim/Overview of the practical:**

The main objective of this practical is to create and understand the working of Pull Requests (PRs) in GitHub for team-based and collaborative development.

2. **Task to be done:**

Create and manage branches in a git repository and synchronize them with a GitHub remote repository to practice collaborative version control.

3. **Steps for Experiment/practical:**

- Open the local Git repository and link it to a remote GitHub repository using the command `git remote add origin <repo-URL>`.
- Create a new branch in the local repository using `git branch <branch-name>` and switch to it using `git checkout <branch-name>` or `git switch <branch-name>`.
- Make the necessary changes in the newly created branch, stage the updated files using `git add`, and commit the changes with an appropriate commit message.
- Push the branch to the remote GitHub repository using `git push -u origin <branch-name>` and confirm that the branch has been successfully uploaded.
- Merge the branch into the main branch either by creating a Pull Request on GitHub or by using the `git merge <branch-name>` command locally, and resolve any merge conflicts if they occur.

4. **Learning outcomes:**

- Collaborative Development Understanding: Learns how pull requests support teamwork in software development.
- Branch Handling in Git: Gains practice in creating and managing branches properly.
- Working with Remote Repositories: Understands how to push code to GitHub successfully.
- Pull Request Process: Learns how to raise, review, and merge pull requests confidently.
- Merge Conflict Handling: Gets exposure to resolving conflicts during merges.
- Real-Time Practical Experience: Gains hands-on experience with a standard industry collaboration workflow.

5. Result: This program has been successfully executed.

6. Output:

```
adity@AdityaSKashyap MINGW64 ~/onedrive/desktop/CollegeCodeSem6/BRM-Sem-6 (main)
$ git switch Exp1/3
Switched to branch 'Exp1/3'
Your branch is up to date with 'origin/Exp1/3'.

adity@AdityaSKashyap MINGW64 ~/onedrive/desktop/CollegeCodeSem6/BRM-Sem-6 (Exp1/3)
$ git status
On branch Exp1/3
Your branch is up to date with 'origin/Exp1/3'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        BRM-1.3-Aditya.pdf

nothing added to commit but untracked files present (use "git add" to track)

adity@AdityaSKashyap MINGW64 ~/onedrive/desktop/CollegeCodeSem6/BRM-Sem-6 (Exp1/3)
$ git add .

adity@AdityaSKashyap MINGW64 ~/onedrive/desktop/CollegeCodeSem6/BRM-Sem-6 (Exp1/3)
$ git commit -m "added 1.3 pdf"
[Exp1/3 90c04f2] added 1.3 pdf
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 BRM-1.3-Aditya.pdf

adity@AdityaSKashyap MINGW64 ~/onedrive/desktop/CollegeCodeSem6/BRM-Sem-6 (Exp1/3)
$ git push
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 249 bytes | 249.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/sundaysheep/BRM-S6.git
aa2354f..90c04f2 Exp1/3 -> Exp1/3

adity@AdityaSKashyap MINGW64 ~/onedrive/desktop/CollegeCodeSem6/BRM-Sem-6 (Exp1/3)
$
```

Exp1/3 - Aditya #1

 **Open** sundaysheep wants to merge 2 commits into `main` from `Exp1/3` 

 Conversation **0**  Commits **2**  Checks **0**  Files changed **3**



sundaysheep commented [now](#)

Owner ...

Merge with main



 sundaysheep added 2 commits [22 minutes ago](#)



 Added 1.3 doc

[aa2354f](#)



 added 1.3 pdf

[90c04f2](#)



No conflicts with base branch

Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions.](#)

Still in progress? [Convert to draft](#)

Exp1/3 - Aditya #1

Merged sundaysheep merged 2 commits into `main` from `Exp1/3` now

Conversation 0 Commits 2 Checks 0 Files changed 3



sundaysheep commented 1 minute ago

Owner ...

Merge with main





sundaysheep added 2 commits 23 minutes ago



Added 1.3 doc

aa2354f



added 1.3 pdf

98c84f2



sundaysheep merged commit 5cba0c7 into `main` now

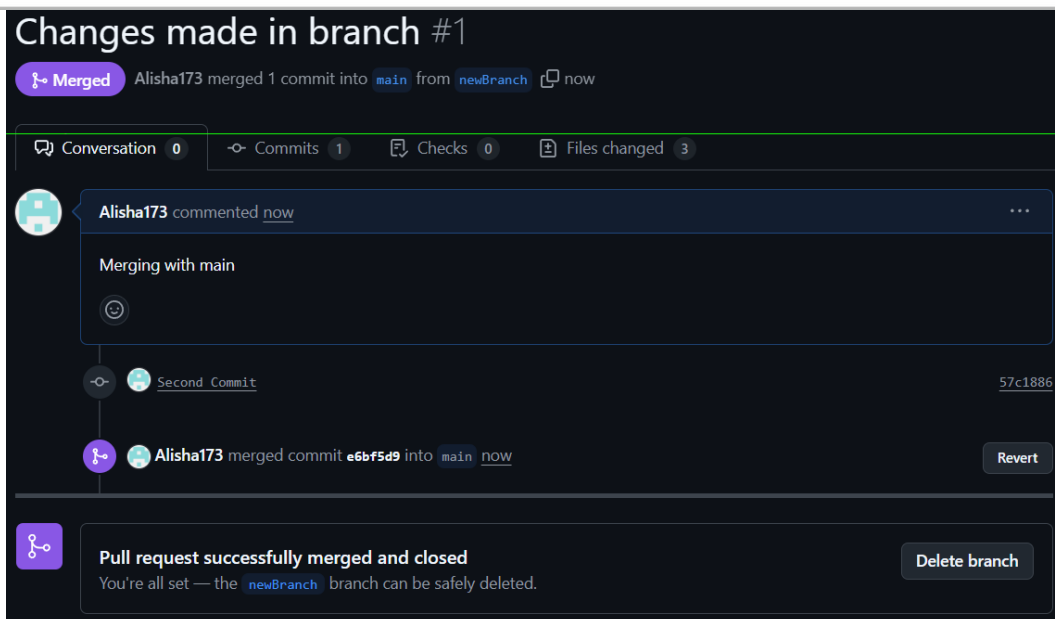
Revert



Pull request successfully merged and closed

You're all set — the `Exp1/3` branch can be safely deleted.

Delete branch



Evaluation Grid (To be created as per the SOP and Assessment guidelines by the faculty):

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.			
2.			
3.			

7. Viva Questions

- What is What is a branch in Git, and why is it used?
- How do you create and switch between branches locally?
- How do you push a branch to a remote GitHub repository?
- What is a Pull Request (PR) in GitHub, and why is it important?
- How do you resolve conflicts during branch merging?

Web Links:

- [Official Git Documentation](#)
- [TutorialsPoint – Git Tutorial](#)